

DRAFT BOX

Match Engine & Tactics Rulebook

How teams actually play against each other — every rating, formula, and formation matchup used by the match simulator, laid out exactly as implemented.

Draft Box v1.1.3 · github.com/DeeplnkGroup/draft-box

1. Overview

Every match in Draft Box — bot vs. bot, human vs. bot, or human vs. human — is decided by the same engine. It has three ingredients:

1. **Player quality.** The overall rating of each of the 11 drafted (or bot-picked) players, weighted by how advanced or deep their exact slot sits on the pitch.
2. **Formation shape.** A tactical fingerprint derived purely from the formation's slot layout — how attack-minded it is, how defensively solid it is, and how wide it stretches the pitch.
3. **Chance.** Goals are sampled from a Poisson distribution around each team's "expected goals" number, so the better team wins more often — not every time.

Nothing here is hardcoded per formation or per team — every number below is derived from the same slot coordinates used to draw the pitch diagram in the app, so the numbers in this document are exactly what the live server computes.

2. The 16 Formations

Every formation is an explicit 11-slot layout (not just a GK/DF/MF/FW count) — this is what lets a player choose exactly which side of defense, midfield, or attack a drafted player fills.

Formation	GK	DF	MF	FW	Style
4-3-3	1	4	3	3	Attacking with width. Three forwards create constant threat.
4-4-2	1	4	4	2	Balanced and reliable. Two banks of four with a strike partnership.
4-2-3-1	1	4	5	1	Defensive solidity with a creative No.10 behind a lone striker.
4-5-1	1	4	5	1	Ultra-compact midfield. Absorb pressure and hit on the counter.
3-4-3	1	3	4	3	High-risk, high-reward. Bold width in attack, exposed at the back.
3-5-2	1	3	5	2	Wing-backs provide the width while a five-man midfield dominates the center.
5-4-1	1	5	4	1	Fortress at the back. Sit deep, stay compact, strike on the break.
4-1-2-1-2	1	4	4	2	A narrow diamond in midfield builds through the center.
4-4-1-1	1	4	4	2	A second striker links midfield and attack just behind the target man.
5-3-2	1	5	3	2	Three at the back become five in defense. Cautious and hard to break down.
3-4-1-2	1	3	5	2	A free-roaming No.10 supports a strike duo behind a solid back three.
4-2-2-2	1	4	4	2	Two holding midfielders shield the back four while two playmakers link the strikers.
5-2-3	1	5	2	3	A back five with only two in the middle — width and firepower up top, but the engine room can get overrun.
5-3-1-1	1	5	3	2	Extreme defensive solidity with a target man and a support striker linking play — built to absorb and break at pace.

Formation	GK	DF	MF	FW	Style
3-2-4-1	1	3	6	1	A back three protected by a double pivot, exploding into four attacking outlets — thrilling going forward, thin at the back.
4-2-4	1	4	2	4	Four forwards, two midfielders. Devastatingly direct but leaves acres of space to exploit in midfield and behind.

3. Formation Shape (the tactical fingerprint)

Three numbers are derived once per formation, straight from its slot list — no per-formation tuning:

```
defShape = (every DF slot) + (every MF slot with y >= 55, i.e. sitting deep)
atkShape = 1.5 × (every FW slot) + (every MF slot with y <= 35, i.e. pushed forward)
width    = (max x) - (min x) among outfield slots
```

(y is pitch depth: 0 = the opponent's goal line, 100 = your own goal / GK. x is 0–100 left-to-right.) A formation with a back five and only one striker (5-4-1) has very high defShape and very low atkShape. A front four with two holding midfielders (4-2-4) is the exact opposite.

Formation	atkShape	defShape	width
4-3-3	4.5	4	70
4-4-2	3.0	4	70
4-2-3-1	4.5	6	70
4-5-1	1.5	5	76
3-4-3	4.5	3	72
3-5-2	3.0	4	80
5-4-1	1.5	5	80
4-1-2-1-2	4.0	5	70
4-4-1-1	3.0	4	70
5-3-2	3.0	5	80
3-4-1-2	4.0	3	72
4-2-2-2	5.0	6	70
5-2-3	4.5	5	80
5-3-1-1	3.0	6	80
3-2-4-1	5.5	5	76
4-2-4	6.0	4	70

4. Player Ratings

Every filled slot gets a continuous attack/defense weight from its own pitch depth (y):

```
attackWeight(slot) = 1 - y / 100  
defenseWeight(slot) = y / 100
```

A striker (y~14) is weighted ~0.86 toward attack and only ~0.14 toward defense. A center back (y~78) is the reverse. A team's two headline numbers are then a weighted average of its 11 players' overall ratings:

```
TeamAttack    = SUM(overall_i × attackWeight_i) / SUM(attackWeight_i) +  
starBonus  
TeamDefense   = SUM(overall_i × defenseWeight_i) / SUM(defenseWeight_i) +  
starBonus
```

Star Bonus. starBonus = min(3, numRealStarPlayersOnTheXI) × 1.5, added to both Attack and Defense. A squad with three or more of the handful of real, well-known stars seeded into the data gets a flat +4.5 rating lift, reflecting composure and quality that a raw overall average under-counts. A fourth star adds nothing further — the bonus caps out.

Captain Bonus (room-optional). When a room has captains enabled, every member picks one of their own 11 players as captain once their draft is complete — the whole room must confirm a captain (if the option is on) before the World Cup can begin. That player's overall is boosted *before* the weighting above, so the bonus counts for more or less depending on how advanced or deep their slot is:

```
effectiveOverall(captain) = min(99, overall + 6)
```

Bots never have a captain — this is a reward for the deliberate choice a human squad-builder makes.

5. Chemistry

Beyond individual quality, how well an XI plays as a *unit* matters. Chemistry is three parameters, each 0–1, averaged into one score:

```
Linkage (L)    = share of "adjacent" pitch partnerships sharing a real source team
Balance (B)    = clamp( 1 - stdDev(overalls) / 20 , 0, 1 )
Leadership (S) = min(1, starPlayersOnXI / 3)
Chemistry      = (L + B + S) / 3
multiplier     = 0.94 + Chemistry × 0.12
```

The multiplier applies to both Attack and Defense, after the captain bonus. A 0.94–1.06 range means chemistry, like the formation edge, is a real but modest swing — it rewards a cohesive XI without letting it override raw quality.

5.1 Linkage — who's "adjacent"? Every formation's slot layout defines a set of local partnerships: two slots within a short distance of each other on the pitch (a center back and the full back beside him, a central midfield trio, and so on — the same x/y coordinates used everywhere else in this document, no separate data). Linkage is the fraction of those partnerships where *both* players were drafted from the same real national team.

5.2 Worked example. An XI built entirely from Brazil's real squad (bots always draw their whole XI from one nation) versus a "franken-XI" of the same average overall (85) assembled from 11 different countries, both in a 4-3-3:

	Linkage	Balance	Leadership	Chemistry	Multiplier
Real national squad	1.00	0.89	1.00	0.96	×1.056
11-country all-star XI	0.00	1.00	1.00	0.67	×1.020

*Simulated 3,000 times at equal overall (85 each): the real national squad won **54%** of the time to the all-star XI's **22%** (rest drawn). This is a deliberate trade-off, not a bug: bot-controlled real countries are guaranteed perfect Linkage, so a human squad built purely by chasing the highest overalls across every country in the draft pool will often be trading chemistry away for star power — both are viable strategies.*

6. Match Simulation

6.1 Formation Edge. How much team X's attacking shape overwhelms team Y's defensive shape, plus a bonus for X being meaningfully wider than Y:

```
edge(X->Y) = clamp( (atkShape_X - defShape_Y) × 0.4 + (width_X - width_Y) × 0.06 , -6, +6 )
```

Worked example (pure width, everything else equal). 4-4-2 and 3-5-2 have *identical* atkShape (3.0) and defShape (4) — the only difference is width (70 vs. 80). Yet:

```
edge(3-5-2 attacking -> 4-4-2 defending) = +0.20  
edge(4-4-2 attacking -> 3-5-2 defending) = -1.00
```

3-5-2's extra width alone is worth a full 1.2-point swing in its favor — stretching a narrower back four creates gaps a flat 4-4-2 doesn't have to worry about defending.

6.2 Expected Goals.

```
xG(X vs Y) = clamp( 1.35 + (Attack_X - Defense_Y) / 18 + edge(X->Y) / 10 , 0.15, 4.5 )
```

Both teams' xG are computed independently and simultaneously (X attacking into Y's defense, and Y attacking into X's defense) — note the formation edge divides by 10 while the base 1.35 and player-quality term dominate the number. This is deliberate: **the tactical matchup nudges the scoreline by roughly half a goal at most; player quality is still the primary driver of most results.**

6.3 Goals. The actual scoreline is sampled from a Poisson distribution using each team's own xG as the mean — the standard way to turn an "expected goals" number into a real, whole-number, occasionally-surprising result.

6.4 Knockout draws -> penalties.

```
composure = Attack + Defense  
P(X wins shootout) = clamp( 0.5 + (composure_X - composure_Y) / 400 , 0.35, 0.65 )
```

Penalties are mostly a coin flip by design (clamped to a 35–65% range) — the better side gets a real but modest edge, never a lock.

7. Formation Matchups

Holding player quality equal, here is every formation's best and worst matchups by pure formation edge (positive = advantage attacking into that opponent's defense). Remember: this is a nudge on top of player quality, not a guarantee — a Brazil XI in a "weak" matchup will still usually beat a Jordan XI in a "strong" one.

Formation	Strong against	Struggles against
4-3-3	3-4-3 (+0.5), 3-4-1-2 (+0.5), 4-4-2 (+0.2)	5-3-1-1 (-1.2), 5-2-3 (-0.8), 5-3-2 (-0.8)

Formation	Strong against	Struggles against
4-4-2	3-4-3 (-0.1), 3-4-1-2 (-0.1), 4-3-3 (-0.4)	5-3-1-1 (-1.8), 5-2-3 (-1.4), 5-3-2 (-1.4)
4-2-3-1	3-4-3 (+0.5), 3-4-1-2 (+0.5), 4-3-3 (+0.2)	5-3-1-1 (-1.2), 5-2-3 (-0.8), 5-3-2 (-0.8)
4-5-1	3-4-3 (-0.4), 3-4-1-2 (-0.4), 4-3-3 (-0.6)	5-3-1-1 (-2.0), 5-2-3 (-1.6), 5-3-2 (-1.6)
3-4-3	3-4-1-2 (+0.6), 4-3-3 (+0.3), 4-4-2 (+0.3)	5-3-1-1 (-1.1), 5-2-3 (-0.7), 5-3-2 (-0.7)
3-5-2	3-4-3 (+0.5), 3-4-1-2 (+0.5), 4-3-3 (+0.2)	5-3-1-1 (-1.2), 5-2-3 (-0.8), 5-3-2 (-0.8)
5-4-1	3-4-3 (-0.1), 3-4-1-2 (-0.1), 4-3-3 (-0.4)	5-3-1-1 (-1.8), 5-2-3 (-1.4), 5-3-2 (-1.4)
4-1-2-1-2	3-4-3 (+0.3), 3-4-1-2 (+0.3), 4-3-3 (+0.0)	5-3-1-1 (-1.4), 5-2-3 (-1.0), 5-3-2 (-1.0)
4-4-1-1	3-4-3 (-0.1), 3-4-1-2 (-0.1), 4-3-3 (-0.4)	5-3-1-1 (-1.8), 5-2-3 (-1.4), 5-3-2 (-1.4)
5-3-2	3-4-3 (+0.5), 3-4-1-2 (+0.5), 4-3-3 (+0.2)	5-3-1-1 (-1.2), 5-2-3 (-0.8), 5-4-1 (-0.8)
3-4-1-2	3-4-3 (+0.4), 4-3-3 (+0.1), 4-4-2 (+0.1)	5-3-1-1 (-1.3), 5-2-3 (-0.9), 5-3-2 (-0.9)
4-2-2-2	3-4-3 (+0.7), 3-4-1-2 (+0.7), 4-3-3 (+0.4)	5-3-1-1 (-1.0), 5-2-3 (-0.6), 5-3-2 (-0.6)
5-2-3	3-4-3 (+1.1), 3-4-1-2 (+1.1), 4-3-3 (+0.8)	5-3-1-1 (-0.6), 5-3-2 (-0.2), 5-4-1 (-0.2)
5-3-1-1	3-4-3 (+0.5), 3-4-1-2 (+0.5), 4-3-3 (+0.2)	5-2-3 (-0.8), 5-3-2 (-0.8), 5-4-1 (-0.8)
3-2-4-1	3-4-3 (+1.2), 3-4-1-2 (+1.2), 4-3-3 (+1.0)	5-3-1-1 (-0.4), 5-2-3 (-0.0), 5-3-2 (-0.0)
4-2-4	3-4-3 (+1.1), 3-4-1-2 (+1.1), 4-3-3 (+0.8)	5-3-1-1 (-0.6), 5-2-3 (-0.2), 5-3-2 (-0.2)

Reading this: very defensive back-five shapes (5-3-1-1, 5-2-3, 5-3-2) are hard for anyone to break down, so they show up as the toughest matchup across the board. Wide, attack-heavy shapes (4-2-4, 3-2-4-1, 5-2-3) punish narrow or thin-on-defense formations (3-4-3, 3-4-1-2) the hardest. Bots are assigned a random formation for every tournament, so these matchups play out differently every run.

8. Tournament Format Recap

- 48 teams -> 12 groups of 4, drawn at random.
- Each group plays a round robin (3 matchdays, 6 matches).
- Top 2 of each group (24) + the 8 best third-placed teams advance to the Round of 32 — the real 2026 FIFA World Cup format.
- Round of 32 -> Round of 16 -> Quarter-Finals -> Semi-Finals -> Final, single elimination.
- Any knockout draw goes to penalties (formula in Section 6.4).

Appendix: Data Disclaimer

The 48-team list matches the real, official field for the 2026 FIFA World Cup. Squad rosters are a mix of a handful of real, well-known players per major team plus procedurally generated names and ratings to fill out playable squads — not an official or licensed dataset. See RULES.md Section 9 for full details.